

Bhushan Ladgaonkar

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PROFESSIONAL SUMMARY

Security Researcher and Machine Learning Engineer specializing in **Adversarial AI** and **AI Security**. Proven experience engineering ML-based Network Intrusion Detection Systems for enterprise infrastructure and building hardware-optimized local LLM inference systems on Apple Silicon. Seeking to leverage expertise in adversarial robustness, simulation, and security architectures.

EDUCATION

Fr. Conceicao Rodrigues College of Engineering
Bachelor of Technology (B.Tech) in Computer Engineering

Mumbai, India
Expected May 2028

EXPERIENCE

Oil and Natural Gas Corporation (ONGC)

Mumbai, India

Summer Intern – Database and Security Group

May 2025 – June 2025

- Developed a prototype **Network Intrusion Detection System (NIDS)** on the UNSW-NB15 dataset (2M+ records) to identify zero-day threats, addressing limitations of traditional signature-based detection.
- Implemented **Random Forest** and **XGBoost** classifiers with **SMOTE**-based class imbalance handling and advanced feature engineering, reducing false positive rates while sustaining high detection sensitivity.
- Collaborated with the Database Group to document threat detection methodologies, translating raw data findings into actionable security intelligence.

TECHNICAL PROJECTS

Gemma-M4-Agent: Local LLM Inference System | *Python, MLX, FastAPI, ChromaDB, React* Apr 2026 – Ongoing

- Built a full-stack local LLM agent around **Google Gemma** on Apple Silicon M4 via **MLX framework** – direct hardware-to-model ownership without wrappers such as Ollama or LM Studio; 8-bit quantized model footprint of **8.5 GB** on Unified Memory.
- Engineered a **Hybrid Memory Architecture** decoupling SSD-resident model weights from USB-persistent **ChromaDB** vector store (384-dim embeddings), enabling zero-copy GPU inference with persistent long-term memory across reboots.
- Implemented **dual-gate stop-token defense**: silicon-level halt via MLX kernel token IDs and software-level control-sequence buffer matching, achieving **100% stop-token accuracy** and preventing hallucinated output beyond semantic boundaries.

TISD-Edu-AI: Local-First RAG Tutor | *Python, MLX, ChromaDB, FastAPI, LLMs*

Apr 2026

- Engineered a privacy-focused on-device **RAG** pipeline using **Microsoft Phi-3 (3.8B)** and **ChromaDB**, achieving **41.42 tokens/sec** at 1.72 s latency with a 9.99 GB footprint on Apple Silicon via MLX.
- Ingested 4,600+ pages of academic PDFs into 6,400+ semantic chunks; validated with a custom evaluation suite achieving **81.1% semantic accuracy** across a 100-sample stress test.

PUBLICATIONS

Benchmarking Resilience: Asymmetric Latent Purification Inspired by Generative Diffusion Bottlenecks *Accepted: May 2026*

First & Corresponding Author. International Journal of Research and Innovation in Applied Science, Mumbai.

- Engineered the **Asymmetric Latent Purifier (ALP)**, a zero-shot structural defense against iterative PGD attacks using an ultra-compact 64-channel stochastic bottleneck and a **ResNet-Lite** backbone.
- Optimized inference to **1.25 ms/sample** on Apple Silicon (MLX) – 100x to 1000x faster than iterative diffusion – recovering CIFAR-10 accuracy to **32.06%** (25.68 dB PSNR) under severe adversarial threat.

TECHNICAL SKILLS

Languages: Python, C, SQL, Java

AI and ML: Machine Learning, Reinforcement Learning, RAG, LLMs, MCTS, PyTorch, MLX, ChromaDB, XGBoost

Security: Network Intrusion Detection (NIDS), Adversarial Robustness, OSINT, Linux (Kali, Ubuntu), Reconnaissance

Developer Tools: Git, FastAPI, React, Node.js, Docker, VS Code, Jupyter Notebooks

LEADERSHIP & INVOLVEMENT

Institution's Innovation Council (IIC) – FRCRCE

Mumbai, India

IPR Activities Coordinator

Apr 2026 – Present

- Coordinating IPR, patent filing, and research commercialization to foster an innovation ecosystem among engineering students.

McKinsey and Company – McKinsey Forward Program

Remote

Selected Participant

May 2026 – Present

- Selected for a competitive program focused on structured problem-solving, adaptability, and digital communication for business-technology integration.

CERTIFICATIONS

CS50 Cybersecurity – Harvard University
Cyber Kill Chain – Cybrary

Offensive Security Operations – Cybrary

OSINT – Cybrary